

Mombasa Computers Inventory Management System

Comprehensive System Report and Business Impact Documentation

Executive Summary

Mombasa Computers Inventory Management System is a fully integrated enterprise inventory solution designed specifically for the operational requirements of Mombasa Computers. The system was developed to transform inventory handling from a manual and error-prone process into a centralized, secure, transparent, and efficient digital environment.

Inventory management is one of the most critical functions within a technology retail business because inventory itself represents a major financial investment. In many organizations, poor inventory tracking often results in misplaced stock, inaccurate records, delayed sales, operational inefficiencies, and product theft.

One of the major challenges identified within Mombasa Computers is inventory loss through theft, untracked movement of products, and lack of accountability regarding stock handling. Traditional methods of inventory tracking create gaps that make it difficult to determine who handled a particular item, where it was moved, or when changes occurred.

This inventory management system directly addresses these challenges by introducing complete tracking, accountability, detailed audit trails, controlled access permissions, and strong security mechanisms.

The system not only improves daily operations but also contributes significantly to increased revenue generation, cost reduction, improved transparency, and overall business growth.

Problem Statement

Before implementation of this system, several operational challenges existed:

Inventory Theft and Loss

Products could disappear without sufficient records showing:

- Who handled the item
- When the item was moved
- Which branch transferred it
- Whether the item was sold or misplaced

As a result:

- Revenue losses occurred
- Accountability became difficult
- Internal investigations consumed significant time
- Staff disputes occasionally emerged

Manual Tracking Challenges

Traditional inventory methods created several issues:

- Slow inventory searches
- Human errors during data entry
- Duplicate records
- Incorrect stock quantities
- Difficult audit procedures
- Poor visibility across branches

Lack of Real-Time Information

Management often lacked immediate information regarding:

- Current stock availability
- Product movement
- Branch inventory levels
- Revenue trends
- Sales performance

This often delayed business decisions.

System Overview

The Mombasa Computers Inventory Management System provides a centralized platform that manages:

- Inventory records
- Sales activities
- Transfers
- Repairs
- User management
- Reporting
- Pricing
- Security monitoring

The system supports multiple branches and multiple user roles while maintaining complete accountability.

Core Functional Areas

The system includes:

Inventory Management

Supports:

- Devices

- Monitors
- Smartboards
- Printers
- Phones
- Accessories
- UPS
- HDDs
- RAM/SSD
- Graphic cards
- Chargers

Features include:

- Product registration
- Bulk Excel import
- QR/Barcode scanning
- Search and filtering
- Quantity updates
- Product categorization

Sales Management

Features include:

- Sales processing
- Invoice generation
- Quotation generation
- Revenue monitoring
- Performance tracking
- Client information management

Transfer Management

The system tracks:

- Product transfers between branches
- Transfer dates
- Transfer origin
- Transfer destination
- Responsible users

Maintenance Management

Tracks:

- Devices under repair

- Fault reports
- Technician updates
- Maintenance history

Theft Prevention and Inventory Protection

One of the most important benefits of this system is preventing inventory theft and reducing inventory losses.

The system introduces multiple layers of accountability that make unauthorized inventory movement significantly more difficult.

Activity Logging

Every activity performed within the system is automatically recorded.

Examples include:

- Adding products
- Updating quantities
- Giving products to sales staff
- Product returns
- Product transfers
- Sales processing
- Deleting records

- Price updates

Each log records:

- User ID
- Username
- Time and date
- Product information
- Action performed

How this helps:

If a product goes missing, management can immediately identify:

- Who handled it
- When they handled it
- What action they performed

This creates personal accountability and discourages theft.

Product Movement Tracking

Products can no longer move without documentation.

For example:

If a laptop moves from warehouse storage to a salesperson:

System records:

- Product serial number
- User receiving item
- Date issued
- Branch
- Status

This prevents situations where products disappear without explanation.

Photo Proof Returns

Returned items require image uploads.

How this helps:

Previously:

A user could claim an item was returned while the item condition remained questionable.

With photo evidence:

- Physical condition is documented
- Disputes reduce significantly
- Fraud attempts become difficult

Branch Transfer Tracking

Every transfer between branches becomes traceable.

Management can determine:

- Which branch sent the item
- Which branch received it
- Transfer dates
- Responsible personnel

This reduces losses during movement.

Role-Based Access Restrictions

Not every employee receives full access.

Users only access features required for their responsibilities.

This minimizes opportunities for unauthorized actions.

Detailed Security Measures

Security within the system is implemented using multiple protection layers.

1. Password Hashing using bcrypt

Definition:

Password hashing converts user passwords into unreadable encrypted values before storing them in the database.

Example:

Instead of storing:

password123

System stores:

\$2y\$10\$Xjdfhsjkdhf8734hfhf....

How it helps:

- Database theft does not expose user passwords
- Passwords cannot easily be reversed
- Prevents unauthorized account access

Business impact:

Protects company accounts and sensitive operational information.

2. SQL Injection Prevention

Definition:

SQL Injection occurs when attackers attempt to manipulate database queries to gain unauthorized access.

Example of dangerous input:

' OR '1'='1

Without protection:

Attackers could:

- View confidential data
- Delete records
- Modify inventory

The system uses:

PDO Prepared Statements

How it helps:

- Separates data from SQL commands
- Prevents malicious database manipulation
- Protects inventory data

Business impact:

Prevents destruction or theft of company information.

3. Cross-Site Scripting (XSS) Protection

Definition:

XSS attacks inject malicious scripts into web pages.

Without protection:

Attackers may:

- Steal sessions
- Redirect users
- Capture information

Protection method:

- Escaping outputs
- Sanitizing user inputs

How it helps:

Prevents malicious code execution.

Business impact:

Protects users and maintains system integrity.

4. Cross-Site Request Forgery (CSRF) Protection

Definition:

CSRF attacks trick users into performing unwanted actions.

Example:

An attacker attempts to force an authenticated user to delete products unknowingly.

Protection:

CSRF session tokens.

How it helps:

- Validates requests
- Prevents unauthorized actions

Business impact:

Protects inventory records from manipulation.

5. Session-Based Authentication

Definition:

After login, users receive temporary session identifiers.

How it helps:

- Verifies authenticated users
- Prevents unauthorized access
- Sessions expire after inactivity

Business impact:

Protects company accounts from misuse.

6. Role-Based Access Control (RBAC)

Definition:

Users receive permissions according to their responsibilities.

Examples:

Super Admin:

- Full access

Sales:

- View products
- Process sales

Technician:

- Repairs and maintenance only

How it helps:

- Limits access to sensitive operations
- Prevents accidental or intentional misuse

Business impact:

Improves accountability and security.

7. Server-Side Validation

Definition:

All submitted information is validated before database storage.

Checks include:

- Empty fields
- Invalid data
- Incorrect formats
- Unexpected inputs

How it helps:

Prevents:

- Invalid records
- Corrupted data
- malicious submissions

Business impact:

Improves database accuracy.

Revenue Generation and Business Growth Impact

The system contributes directly and indirectly toward increased company revenue.

Reduced Product Theft

Inventory theft directly reduces profit.

If products worth KSh 300,000 disappear annually due to weak tracking:

Reducing losses by even 70% results in:

Annual savings:

KSh 210,000+

Improved Stock Visibility

Sales teams can immediately determine:

- Available products
- Product location

- Quantity remaining

Benefits:

- Faster customer response
- Reduced lost sales opportunities
- Better customer satisfaction

Faster Sales Transactions

Features such as:

- QR scanning
- Quick search
- Automated quotations
- Invoice generation

Reduce customer waiting time.

Result:

More transactions completed daily.

Estimated increase:

15–20% more processed sales.

Better Pricing Control

Centralized pricing prevents:

- Underpricing
- Overpricing
- Inconsistent branch prices

Result:

Improved profit margins.

Estimated increase:

5–10% profit improvement.

Better Stock Distribution

Transfer management allows inventory movement based on demand.

Example:

If printers sell faster in MOI branch than KIMATHI branch:

Management can transfer stock quickly.

Benefits:

- Reduced idle stock
- Improved sales opportunities

Better Decision-Making

Management reports provide insights on:

- Top-selling products
- Low-performing products
- Revenue trends
- Staff performance

Benefits:

- Better purchasing decisions

- Better investment planning

Estimated Return on Investment (ROI)

Time Savings

Activity | Estimated Annual Hours Saved

Manual stock entry | 500+

Inventory searching | 250+

Audit preparation | 300+

Dispute investigations | 200+

Reporting | 150+

Total:

1,400+ hours annually

Financial Benefits

Estimated improvements:

- Reduced inventory loss: KSh 200,000–500,000 annually

- Increased sales efficiency: 15–20%
- Increased profit margins: 5–10%
- Reduced operational costs

Conclusion

The Mombasa Computers Inventory Management System is more than an inventory application; it is a complete business management and protection platform.

The system transforms inventory operations through:

- ✓ Theft prevention
- ✓ Complete accountability
- ✓ Strong security architecture
- ✓ Real-time visibility
- ✓ Revenue optimization
- ✓ Operational efficiency
- ✓ Accurate reporting

✓ Business growth support

By introducing transparency and control over inventory activities, the system minimizes losses while maximizing profitability and productivity.

The implementation of this system positions Mombasa Computers for sustainable growth and improved operational performance in a competitive business environment.